TAB-PO: Preference Optimization with a Token-Level Adaptive Barrier for Token-Critical Structured Generation

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Abstract

Knowledge-based systems increasingly use large language models (LLMs) for ontology-conformant knowledge extraction and structured prediction from unstructured text, involving semantic labelling, relational linking, and textual grounding. After supervised fine-tuning (SFT), residual errors often occur at ontology boundaries, where confusable labels, competing evidence spans, and plausible alternative links differ by only a few decisive tokens. Direct Preference Optimization (DPO) is poorly suited to such outputs: sequence-level updates are diluted by shared serialization tokens (*gradient dilution*) and may reduce the likelihood of rare, correct ontology tokens (*preferred-token erosion*). We propose a knowledge-guided preference-learning framework with two components. Confusion-aware preference construction combines expert-curated ambiguity patterns with observed SFT errors to generate minimally perturbed, schema-valid negatives. Token-Adaptive Barrier Preference Optimization (TAB-PO) adds a confidence-gated token-level likelihood barrier to protect under-confident preferred ontology tokens. We evaluate six Llama

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and Qwen models (1.5B–70B) on clinical patient–provider communication extraction (PV-Miner) and scientific knowledge-graph extraction (SciERC). On SciERC, the pooled mean Entity, Relation, and Coreference F1 across six models improves by 11.59% relative to SFT. Across both tasks, TAB-PO exceeds the strongest of six DPO-family baselines in 39 of 42 model–metric cells, including all 30 ontology-critical cells. TAB-PO Qwen2.5-14B achieves 71.23 four-metric macro F1, 4.24 points above the composite of frontier per-metric maxima; Llama-3.2-3B also surpasses nine of 11 frontier models in macro F1. Llama-3.3-70B exceeds the frontier per-metric composite over Entity, Relation, and Coreference by 14.71% relative.

Keywords: Knowledge extraction, Ontology-constrained structured prediction, Preference optimization, Large language models, Clinical communication mining, Scientific knowledge graphs

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1. Introduction

Knowledge-based systems increasingly depend on large language models (LLMs) to turn unstructured text into structured, ontology-conformant knowledge. This is a structured prediction problem: the output is not free text but a set of interdependent records, and it is useful only if every semantic label, relational link, and evidence span in it is correct. Clinical registries, scientific knowledge graphs and decision-support pipelines all need the same kind of output: records whose labels come from a fixed ontology, whose relations respect its constraints, and whose every assertion is grounded in evidence from the source text [1, 2]. Errors in this conversion flow directly into the knowledge base. In clinical text they produce missed or wrongly normalized findings [3]; in legal and regulatory review they introduce compliance risk [4]; and in biomedical extraction a wrong label or link injects noise that downstream reasoning inherits [5]. Because downstream systems treat these records as facts, errors in extraction and structured prediction propagate into every high-stakes application built on them [6].

Supervised fine-tuning teaches the model the target schema and valid output format, but many residual errors occur at hard decision boundaries between plausible schema-valid alternatives, such as closely related ontology labels, competing evidence spans, or alternative relational links [2, 7]. Preference optimization is therefore a natural post-SFT stage: it can contrast ambiguous correct and incorrect structured outputs and sharpen the model's

choices [8, 9].

Direct Preference Optimization (DPO) has become a widely used approach for post-supervised alignment of large language models, replacing explicit reward modeling and online reinforcement learning with an offline Bradley–Terry objective that increases the relative likelihood of preferred responses over rejected responses [9]. DPO and its variants have been effective for open-ended generation tasks such as instruction following, dialogue, and summarization [8, 10–13]. Ontology-driven structured prediction, however, has a different error geometry. Preferred and rejected outputs are often near-identical serialized records that differ in only a few schema-defining tokens while sharing most JSON scaffolding, field names, and formatting tokens.

This low-separation regime exposes a mismatch between sequence-level preference objectives and token-critical structured correctness. Standard DPO can dilute learning signal across non-critical serialization tokens rather than concentrating it on the sparse tokens that determine semantic labels, textual groundings, or relational links; we refer to this mismatch as *gradient dilution*. It can also improve the aggregate preferred-over-rejected margin while reducing the likelihood of under-confident preferred schema tokens, a failure mode we call *preferred-token erosion*. Recent DPO variants address sequence-level pathologies such as calibration drift, reference dependence, or positive-likelihood degradation [14–17], and token-level variants introduce finer-grained preference signals [18, 19]. However, these methods do not explicitly target the low-edit-distance, ontology-constrained preferences that arise in structured extraction.

Addressing this regime requires preference pairs that reflect realistic ontology-level errors and an objective that focuses optimization on the few tokens that decide structured correctness while still improving the preferred-over-rejected preference margin. Our contributions are as follows:

- **We introduce a general task formulation for ontology-driven structured prediction.** We define a task-agnostic structured-output setting in which correctness depends on semantic labelling, textual grounding, and relational linking under a task-specific ontology (Section 3).
- **We mathematically characterize why sequence-level DPO is mismatched to low-edit-distance structured preferences.** Through a token-level decomposition of the DPO gradient, we show that standard DPO can allocate update mass to non-critical serialization tokens rather than the sparse tokens that determine structured correctness, leading to *gradient*

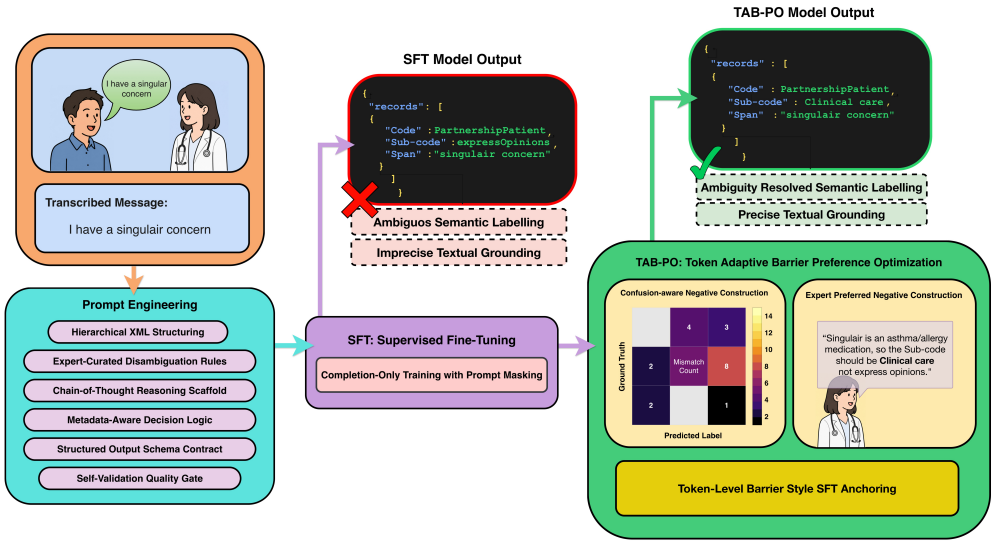


Figure 1: Overview of the TAB-PO framework for ontology-driven structured prediction. A modular prompt-engineered interface and completion-only supervised fine-tuning teach the model to generate schema-formatted structured records, but residual SFT errors can remain in token-critical decisions such as semantic labelling, relational linking and textual grounding. TAB-PO targets these errors using confusion-aware and expert-guided hard negatives together with a confidence-gated token-level SFT-style likelihood barrier, yielding outputs with resolved semantic and relational ambiguity and more precise source grounding.

dilution. We further show that the aggregate preferred-over-rejected margin can improve even when the likelihood of rare or under-confident preferred schema tokens decreases, causing *preferred-token erosion* (Appendix A of the Supplementary Material).

- **We construct confusion-aware structured preference pairs.** We create minimally perturbed, schema-valid hard negatives from empirical SFT ontology-level confusions and expert-adjudicated ambiguity patterns, targeting realistic semantic-label, evidence-grounding, and relational-linking errors (Section 6).
- **We propose Token-Adaptive Barrier Preference Optimization.** TAB-PO augments preference optimization with a confidence-gated SFT-style likelihood regularizer on under-confident preferred tokens. Together with confusion-aware preferences, TAB-PO reduces gradient dilution, protects vulnerable ontology-bound tokens from likelihood erosion, and preserves the structural scaffolding required for valid serialized outputs (Sec-

tion 7).

- **We provide comprehensive empirical and diagnostic evidence.** Across PV-Miner and SciERC, and across Llama and Qwen models from 1.5B to 70B parameters, TAB-PO consistently improves over SFT and outperforms strong sequence-level and token-level DPO-family baselines. Mechanistic diagnostics further substantiate these gains by showing that TAB-PO concentrates optimization on schema-critical tokens while maintaining preference separation (Section 8; Section 7.1).

Figure 1 provides an overview of the TAB-PO framework, showing how the modular prompt-engineered interface, supervised structured initialization, confusion-aware preference construction, and token-adaptive barrier objective work together for ontology-driven structured prediction.

2. Related Work

TAB-PO connects structured information extraction with post-SFT preference learning. We situate its contribution around four challenges: distinguishing schema-valid interpretations, preserving correct-token likelihood, providing direct token-level supervision, and constructing informative structured negatives.

Structured extraction and post-SFT alignment. Joint extraction models capture dependencies among entities, relations, and coreference [2]. UIE unifies extraction through a structured generation language, while GenIE constrains generated triplets to a knowledge-base schema [20, 21]. GoLLIE incorporates annotation guidelines, and grammar-constrained decoding improves the validity of clinical extraction outputs [1, 7]. ADELIE extends extraction alignment beyond SFT by applying DPO to gold-versus-generated answers and sampled answers ranked by BLEU [22]. A remaining challenge is to focus post-SFT learning on sparse ontology-level mistakes that distinguish otherwise schema-admissible records. TAB-PO addresses this challenge by coupling preferences built around specific structured errors with selective restoration of low-confidence preferred tokens.

Preference objectives and likelihood preservation. DPO learns from completion pairs without an explicit reward model or online policy rollouts [9]. IPO regularizes preference margins, SimPO uses a reference-free, length-normalized reward, and Cal-DPO calibrates implicit reward magnitudes [14–16]. DPOP

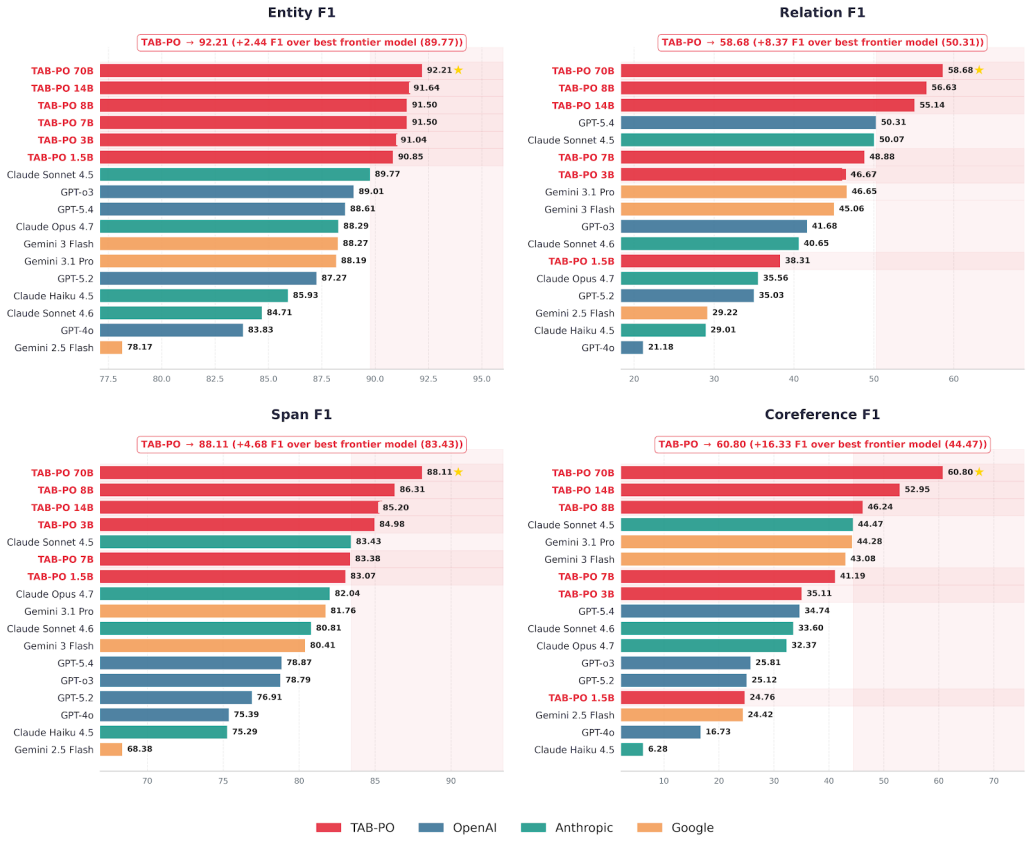


Figure 2: SciERC F1 comparison between TAB-PO models and frontier models across Entity, Relation, Span, and Coreference extraction. TAB-PO 70B achieves the best performance across all four evaluation dimensions, with particularly large gains on Relation F1 and Coreference F1, the most structurally demanding SciERC tasks. These results demonstrate the effectiveness of token-adaptive barrier preference optimization for ontology-constrained structured prediction.

identifies preferred-likelihood degradation on low-edit-distance pairs and adds a reference-relative completion-level penalty [17]; likelihood-displacement analysis links this behavior to similarity between preferred and rejected representations [23]. SLiC-HF and CPO incorporate supervised regularization [24, 25], while APO controls likelihood changes according to the relationship between the model and preference data [26]. However, aggregate likelihood control does not ensure preservation of each correct token, and uniformly applied supervision does not select positions according to their current confidence. TAB-PO localizes its additional supervised loss to low-confidence preferred

positions, providing a direct restoration signal even when the completion-level preference margin improves.

Token-level preference learning. TDPO introduces token-wise forward-KL constraints; TI-DPO combines attribution-based importance with a triplet-style loss; and TIS-DPO estimates token weights from contrastive models [18, 19, 27]. Two particularly close precedents use policy confidence: ConfPO selects low-confidence tokens for preference scoring [28], and BMC upweights differing tokens in correlated response pairs within the preference objective [29]. These mechanisms refine token-level preference learning, but leave a distinct design question: how to provide preferred-token supervision independently of the contrastive score. This matters when selected shared-prefix tokens have canceling contributions to the unweighted DPO margin. TAB-PO retains the full reference-adjusted margin and adds an asymmetric, confidence-gated negative-log-likelihood term outside the contrastive score, supplying direct supervision at selected preferred positions even when their margin contributions cancel.

Preference construction and scope. CLAIR uses minimally contrastive revisions, while BMC constructs correlated pairs through targeted response modification [26, 29]. Such revisions establish informative contrasts, but do not by themselves specify which ontology-level alternatives reflect a particular extraction model's recurring errors. TAB-PO uses empirical SFT ontology confusions and expert-observed ambiguities to construct schema-admissible alternatives involving labels, record membership, and relational links. Its contribution is to couple these task-informed negatives with selective preferred-token restoration, linking the structured errors exposed by the data to the optimization applied to preferred outputs. The following formulation makes semantic labelling, textual grounding, and relational linking explicit. Appendix F of the Supplementary Material extends the review with detailed objective comparisons, autoregressive overlap, preference-data design, and verifier-based reinforcement learning.

3. General Task Formulation for Ontology-Driven Structured Prediction

We introduce a general formulation for ontology-driven structured prediction, covering tasks in which the desired output is not free-form text but a schema-valid set of records constrained by a task-specific ontology. Given an

input sequence s, optional task metadata m, and an ontology $\mathcal{O}$, the goal is to predict a finite structured output $\widehat{Y} = \{y_1, y_2, \ldots, y_N\}$. The generation model f_θ maps

$$f_\theta : (s, m, \mathcal{O}) \longmapsto \widehat{Y} = \{y_1, y_2, \ldots, y_N\}. \tag{1}$$

Each predicted record has the abstract form $y_i = (\ell_i, \Pi_i, R_i)$, where the three components correspond to semantic labelling, textual grounding, and relational linking.

Semantic label. The semantic-label component ℓ_i is drawn from the ontology-defined label space. Depending on the task, ℓ_i may denote an entity type, a relation type, or a hierarchical label tuple. For hierarchical ontologies, we write $\ell_i = (\kappa_i, \varsigma_i)$, where the child label ς_i must be valid under the ontology mapping $\mathcal{H}$, i.e., $\varsigma_i \in \mathcal{H}(\kappa_i)$.

Textual grounding. Let $\mathcal{S}(s)$ denote the set of admissible evidence spans in the input sequence. The grounding component $\Pi_i \subseteq \mathcal{S}(s)$ contains the span or spans that anchor record y_i to the source text.

Relational linking. The relational-linking component R_i stores optional typed links from record y_i to other records or grounded mentions. These links encode task-specific inter-record structure, such as relation arguments, directionality, coreference clusters, or other ontology-defined constraints. If a task does not require relational links, then $R_i = \emptyset$.

The ontology $\mathcal{O}$ determines which semantic labels, grounding spans, and relational links are valid. Concrete task-specific instantiations of this general formulation for PV-Miner and SciERC are presented in Appendix D of the Supplementary Material.

4. Modular Prompt Engineering for Structured Prediction

We next introduce a modular prompt-engineering framework for ontology-constrained structured prediction. The framework is instantiated using replaceable task-specific placeholders, allowing the same prompt architecture to be adapted to any structured prediction task. It targets common failure modes of in-context structured generation, including format drift, label ambiguity, reasoning shortcuts, evidence hallucination, metadata-conditioned confusion, and invalid relational links.

The prompt decomposes structured prediction into six explicitly controlled modules. **M1** (XML structuring) partitions global instructions and task-specific constraints into semantically tagged blocks that establish scope boundaries for role, performance target, task definition, ontology, and output requirements [30]. **M2** (disambiguation rules) encodes expert-derived decision boundaries for confusable ontology labels, grounding boundaries, and relational-link validity [7, 31]. **M3** (reasoning scaffold) defines a structured verification routine that orders context analysis, candidate decomposition, semantic labelling, textual grounding, relational-link validation, and final consistency checking before output emission [32]. **M4** (metadata-aware decision logic) converts task metadata into explicit control variables that restrict the valid search space for labels, evidence spans, and links [8, 33]. **M5** (schema contract) declares machine-parseable validity constraints for records, labels, groundings, relational links, and task-specific coverage requirements [7, 34]. **M6** (quality gate) defines a single-turn pre-emission audit for parseability, ontology validity, grounding fidelity, relational-link consistency, disambiguation-rule compliance, and expert defensibility [35, 36].

Appendix C of the Supplementary Material provides the detailed component-level explanation of the modular prompt-engineering framework, and Figure C1 of the Supplementary Material presents the corresponding general-purpose template.

5. Supervised Structured Initialization

After defining the prompt-engineered structured prediction interface, we perform an initial supervised adaptation stage in which the model learns to emit the corresponding gold schema-valid output under this interface. All subsequent preference-optimization stages build on the resulting supervised model p_{SFT}, which is kept fixed as the reference policy.

Given a gold record set $Y = \{y_1, \ldots, y_N\}$, we serialize it into a JSON completion string Y_s and concatenate it with the instantiated prompt x, where x contains the prompt-engineered instruction populated with the input sequence s, and optional metadata m. Let $z_{1:T} = \mathrm{Tok}(x \,\|\, Y_s)$ denote the tokenized causal-LM training sequence, where $\mathrm{Tok}(\cdot)$ is the model tokenizer and $\|$ denotes string concatenation. Following completion-only instruction tuning, we compute loss only over tokens belonging to the serialized completion Y_s:

$$\mathcal{L}_{\mathrm{SFT}}(\theta) = -\mathbb{E}_{(x,Y_s)\sim\mathcal{D}_{\mathrm{SFT}}}\left[\frac{\sum_{t=1}^{T} r_t \log p_\theta(z_t \mid z_{<t})}{\sum_{t=1}^{T} r_t}\right]. \tag{2}$$

where $r_t \in \{0,1\}$ indicates whether token z_t belongs to the completion Y_s. The resulting model p_{SFT} serves as the initialization for preference optimization and is kept fixed as the reference policy in TAB-PO.

6. Confusion-Aware Preference Construction

Having obtained the supervised model p_{SFT}, we next construct preference pairs that expose the model to realistic structured prediction errors. Preference optimization is most effective when the rejected completion represents a plausible failure mode rather than an arbitrary corrupted output. In ontology-driven structured generation, residual SFT errors are often *structured confusions*: the model produces a schema-valid output that is mostly correct but substitutes one plausible ontology-bound decision for another, such as a confusable label, an imprecise evidence span, or an incorrect relational link. These errors are especially challenging because the preferred and rejected outputs may share nearly all serialization tokens and differ only at a small number of task-critical decision points. We therefore construct preference triples (x, Y^+, Y^-), where Y^+ is the gold structured output and Y^- is a minimally perturbed, schema-valid alternative designed to isolate a realistic ontology-level error.

Our construction follows a hybrid protocol. First, we include expert-curated preferences that capture ambiguity patterns identified during annotation review and adjudication (40%). Second, we generate confusion-aware synthetic preferences from empirical errors made by the SFT model on a held-out validation split (60%). This design ensures that preference optimization focuses on realistic ontology-level confusions rather than uninformative random negatives.

Empirical confusion extraction on validation data. Using the supervised model p_{SFT}, we perform inference on a held-out validation set and compare each predicted structured output $\widehat{Y}$ with the gold structured output Y. Because semantic-label confusions are meaningful only when the corresponding records refer to the same textual evidence, we first align gold and predicted records using relaxed span-level matching. For a predicted span $\hat{\pi} \in \widehat{\Pi}_k$ and a gold span $\pi \in \Pi_j$, let $\mathrm{Tok}(\cdot)$ denote the set of tokens in a span. We use a relaxed

span-matching criterion that treats two spans as matched when either span token set contains the other, or when their partial token overlap exceeds a threshold. For the latter case, we compute token-level Jaccard similarity as

$$\text{Jaccard}(\hat{\pi}, \pi) = \frac{|\,\text{Tok}(\hat{\pi}) \cap \text{Tok}(\pi)|}{|\,\text{Tok}(\hat{\pi}) \cup \text{Tok}(\pi)|}. \tag{3}$$

Given a relaxed matching threshold γ, we define

$$\begin{aligned}\text{Match}_{\Pi,\gamma}(\hat{\pi}, \pi) = \mathbb{I}\big[&\text{Tok}(\pi) \subseteq \text{Tok}(\hat{\pi}) \;\vee\; \text{Tok}(\hat{\pi}) \subseteq \text{Tok}(\pi) \\ &\vee\; \text{Jaccard}(\hat{\pi}, \pi) \geq \gamma\big].\end{aligned} \tag{4}$$

Using this span-matching criterion, we form the set of matched gold–predicted record pairs:

$$\mathcal{M}_\gamma(Y, \widehat{Y}) = \{(j,k) : \exists\, \pi \in \Pi_j, \exists\, \hat{\pi} \in \widehat{\Pi}_k \text{ s.t. } \text{Match}_{\Pi,\gamma}(\hat{\pi}, \pi) = 1\}. \tag{5}$$

In our implementation, we set $\gamma = 0.6$. We then collect mismatched semantic label pairs only over these span-aligned records:

$$\mathcal{A}^{\neq}(Y, \widehat{Y}) = \{(\ell_j, \widehat{\ell}_k) :\ (j,k) \in \mathcal{M}_{0.6}(Y, \widehat{Y}), \ell_j \neq \widehat{\ell}_k\}. \tag{6}$$

Aggregating these mismatches over the validation set yields a frequency-weighted empirical confusion distribution over ontology labels.

Preference triples construction. The empirical confusion distribution provides a task-specific estimate of the model's most likely ontology-level failure modes. We therefore use it to guide negative construction, ensuring that rejected outputs remain schema-valid while targeting realistic low-separation errors. Each preference triple shares the same input x and differs only in the structured output. We set $Y^+ = Y$ and construct a minimally perturbed Y^- that preserves schema validity while introducing a realistic structured error. We use four different perturbation families:

1. **Semantic confusion with preserved grounding.** This perturbation changes one ontology label while preserving the grounding and relational structure of the selected record.

   Step 1: estimate label vulnerability. Let $C(\ell, \tilde{\ell})$ denote the number of times the SFT model predicts $\tilde{\ell}$ when the gold semantic label is ℓ on

the validation set. We define the total confusion count for label ℓ as

$$c(\ell) = \sum_{\ell' \neq \ell} C(\ell, \ell'). \tag{7}$$

Step 2: select a vulnerable record. For a structured output $Y^+ = \{y_1^+, \ldots, y_N^+\}$ with $y_i^+ = (\ell_i, \Pi_i, R_i)$, we sample a record index from the vulnerability-weighted distribution

$$q(i \mid Y^+) = \frac{c(\ell_i) + \epsilon}{\sum_{j=1}^{N} (c(\ell_j) + \epsilon)}, \tag{8}$$

where $\epsilon > 0$ provides smoothing. This assigns higher probability to records whose labels are more frequently confused by p_{SFT}.

Step 3: sample a confusable replacement label. Given the selected record $y_i^+ = (\ell_i, \Pi_i, R_i)$, we sample an ontology-valid replacement label $\tilde{\ell}_i$ from the empirical confusion distribution

$$p_{\mathrm{conf}}(\tilde{\ell} \mid \ell_i) = \frac{C(\ell_i, \tilde{\ell}) + \epsilon}{\sum_{\ell'} (C(\ell_i, \ell') + \epsilon)}. \tag{9}$$

Step 4: construct the rejected output. We preserve the grounding and relational components and define $\tilde{y}_i = (\tilde{\ell}_i, \Pi_i, R_i)$. The rejected output is

$$Y^- = Y^+ \setminus \{y_i^+\} \cup \{\tilde{y}_i\}. \tag{10}$$

Thus, the preferred and rejected outputs differ in a single semantic decision while retaining the same evidence spans and relational links.

2. **Missing-record perturbation.** We delete a record from the preferred output to simulate under-extraction:

$$Y^- = Y^+ \setminus \{y_i^+\}, \tag{11}$$

where y_i^+ is selected according to Eq. 8.

3. **Extra-record perturbation.** We insert a spurious but schema-valid record to simulate over-extraction:

$$Y^- = Y^+ \cup \{\tilde{y}\}, \tag{12}$$

where $\tilde{y} = (\tilde{\ell}, \tilde{\Pi}, \tilde{R})$ is constructed using an ontology-valid semantic label $\tilde{\ell}$, a candidate grounding set $\tilde{\Pi}$ sampled from the source sequence s, and optional relational links $\tilde{R}$, subject to the task-specific ontology, relational-linking rules, and schema-validity constraints.

4. **Relational-link perturbation.**

   We modify a relational link to simulate relational-linking confusion:

$$Y^- = Y^+ \setminus \{y_i^+\} \cup \{\tilde{y}_i\}, \tag{13}$$

   where $y_i^+ = (\ell_i, \Pi_i, R_i)$ is selected according to Eq. 8, and $\tilde{y}_i = (\ell_i, \Pi_i, \tilde{R}_i)$ preserves the semantic label and grounding spans while replacing R_i with a schema-valid but incorrect relational-link set $\tilde{R}_i$.

For intuitive examples of how these perturbation families produce low-separation, schema-valid hard negatives, see the PV-Miner and SciERC illustrations in Appendix G of the Supplementary Material.

7. Token-Adaptive Barrier Preference Optimization

Motivation. The preference construction described above yields low-separation structured preference pairs, where preferred and rejected completions often differ in only a small number of schema-defining decisions while sharing most of the same serialized output. Standard sequence-level DPO is poorly matched to this regime. First, its gradients can be diluted across JSON scaffolding and non-critical serialization tokens rather than concentrated on the sparse tokens that determine structured correctness. Second, because DPO optimizes only a relative sequence-level margin, it can improve the aggregate preference margin while reducing the likelihood of some correct preferred tokens. We refer to these effects as *gradient dilution* and *preferred-token erosion*. Appendix A of the Supplementary Material provides the mathematical intuition behind these two failure modes and derives the corresponding token-level decomposition of the DPO gradient.

Objective. Let Y_s^+ and Y_s^- denote the serialized preferred and rejected completion strings obtained from Y^+ and Y^-. We define the serialized preference dataset as $\mathcal{D}_{\text{pref}}$, whose elements are (x, Y_s^+, Y_s^-). For each serialized completion Y_s, let $u = \text{Tok}(Y_s) = (u_1, \ldots, u_T)$ denote its completion-token sequence.

We write the completion log-likelihood as

$$\mu_\theta(Y_s \mid x) \triangleq \sum_{t=1}^{T} \log p_\theta(u_t \mid x, u_{<t}). \tag{14}$$

Here, μ_{SFT} denotes the same completion log-likelihood computed under the fixed supervised reference model p_{SFT}. For each preference triple (x, Y_s^+, Y_s^-), we define the reference-adjusted advantage as the difference between the policy log-likelihood margin and the corresponding SFT-reference log-likelihood margin:

$$\Delta_\theta \triangleq \left[\mu_\theta(Y_s^+ \mid x) - \mu_\theta(Y_s^- \mid x)\right] - \left[\mu_{\mathrm{SFT}}(Y_s^+ \mid x) - \mu_{\mathrm{SFT}}(Y_s^- \mid x)\right], \tag{15}$$

The corresponding preference loss is

$$\mathcal{L}_{\mathrm{pref}}(\theta) = -\mathbb{E}_{\mathcal{D}_{\mathrm{pref}}}\left[\log \sigma\left(\beta \Delta_\theta\right)\right]. \tag{16}$$

To address gradient dilution and preferred-token erosion, TAB-PO adds a confidence-gated token barrier on the preferred completion. The barrier activates only when the current policy is under-confident on preferred completion tokens, according to a probability threshold $\tau \in (0, 1)$.

For the preferred serialized completion Y_s^+, let $u^+ = \mathrm{Tok}(Y_s^+) = (u_1^+, \ldots, u_{T^+}^+)$. We define the per-token gate as

$$g_t^\theta(x, u^+) \triangleq \mathbb{I}\left[\log p_\theta(u_t^+ \mid x, u_{<t}^+) < \log \tau\right]. \tag{17}$$

The gate identifies preferred-token positions whose likelihood falls below the confidence threshold. TAB-PO then applies an SFT-style likelihood-restoration penalty only to these activated positions, averaging the preferred-token negative log-likelihood over the selected tokens:

$$\mathcal{L}_{\mathrm{barrier}}(\theta) = \mathbb{E}_{(x, Y_s^+, Y_s^-) \sim \mathcal{D}_{\mathrm{pref}}}\left[\frac{\sum_{t=1}^{T^+} g_t^\theta(x, u^+)\left[-\log p_\theta(u_t^+ \mid x, u_{<t}^+)\right]}{\max\left(1, \sum_{t=1}^{T^+} g_t^\theta(x, u^+)\right)}\right]. \tag{18}$$

When no preferred token is selected by the gate, the numerator is zero and the barrier contributes no loss. In implementation, the binary gate g_t^θ is detached from the computation graph; gradients are propagated only through the selected preferred-token negative log-likelihood terms.

Algorithm 1 Token-Adaptive Barrier Preference Optimization

Require: SFT data $\mathcal{D}_{\text{SFT}}$, validation data $\mathcal{D}_{\text{val}}$, expert preferences $\mathcal{D}_{\text{expert}}$, ontology $\mathcal{O}$, relaxed span-matching threshold γ, smoothing constant ϵ, preference temperature β, confidence threshold τ, barrier weight λ

Ensure: TAB-PO policy p_θ

Train p_{SFT} on $\mathcal{D}_{\text{SFT}}$ using completion-only supervised loss $\mathcal{L}_{\text{SFT}}$.

Initialize $p_\theta \leftarrow p_{\text{SFT}}$ and keep p_{SFT} fixed as the reference model.

Confusion-aware preference construction

Run p_{SFT} on $\mathcal{D}_{\text{val}}$, align records with $\text{Match}_{\Pi,\gamma}$, and estimate confusion statistics $C(\ell, \tilde{\ell})$, $c(\ell)$, and $p_{\text{conf}}(\tilde{\ell} \mid \ell)$.

Using $C(\ell, \tilde{\ell})$, $c(\ell)$, and $p_{\text{conf}}(\tilde{\ell} \mid \ell)$, construct $\mathcal{D}_{\text{pref}}$ by combining $\mathcal{D}_{\text{expert}}$ with schema-valid hard negatives generated from semantic-confusion, missing-record, extra-record, and relational-link perturbations.

TAB-PO preference optimization

for each minibatch $\mathcal{B} \subset \mathcal{D}_{\text{pref}}$ **do**

For each preferred completion Y_s^+, compute the detached confidence gate

$$g_t^\theta(x, u^+) = \mathbb{I}\left[\log p_\theta(u_t^+ \mid x, u_{<t}^+) < \log \tau\right].$$

Update p_θ by minimizing the minibatch TAB-PO objective

$$\widehat{\mathcal{L}}_{\text{TAB-PO}}(\theta; \mathcal{B}) = \frac{1}{|\mathcal{B}|} \sum_{(x, Y_s^+, Y_s^-) \in \mathcal{B}} \Bigg[- \log \sigma\Big(\beta\big(\mu_\theta(Y_s^+ \mid x) - \mu_\theta(Y_s^- \mid x) - \mu_{\text{SFT}}(Y_s^+ \mid x) + \mu_{\text{SFT}}(Y_s^- \mid x)\big)\Big) + \lambda \frac{\sum_{t=1}^{T^+} g_t^\theta(x, u^+) \left[-\log p_\theta(u_t^+ \mid x, u_{<t}^+)\right]}{\max\left(1, \sum_{t=1}^{T^+} g_t^\theta(x, u^+)\right)} \Bigg].$$

end for

return p_θ

The final TAB-PO objective combines reference-adjusted preference learning with the confidence-gated supervised restoration term:

$$\mathcal{L}_{\text{TAB-PO}}(\theta) = \mathcal{L}_{\text{pref}}(\theta) + \lambda \mathcal{L}_{\text{barrier}}(\theta), \tag{19}$$

where $\lambda \geq 0$ controls the strength of the barrier. Algorithm 1 provides a

compact summary of the TAB-PO procedure, showing how validation-set confusion statistics and expert preferences are used to construct schema-valid hard negatives, which are then optimized with the confidence-gated token-adaptive barrier objective.

7.1. Mechanistic Diagnostics

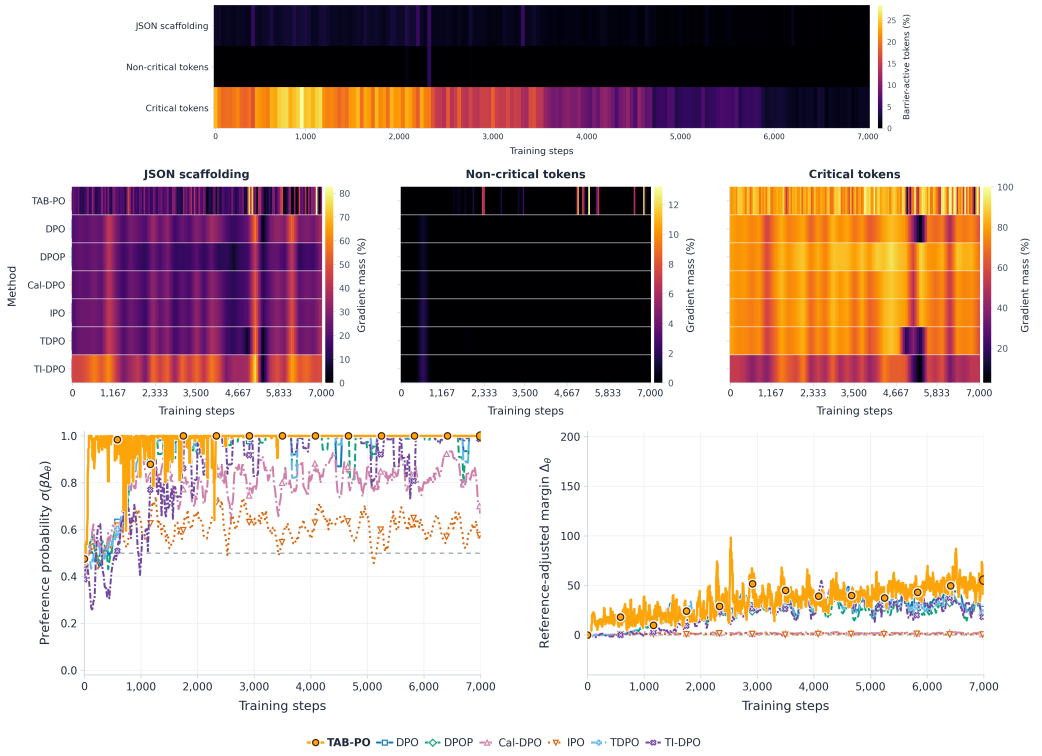


Figure 3: Mechanistic diagnostics for TAB-PO and preference-optimization variants. The top panel shows TAB-PO barrier activity across JSON scaffolding, non-critical serialization tokens, and critical schema tokens. The middle panel shows gradient-mass allocation across the same token categories for TAB-PO and DPO-family baselines. The bottom panel shows preference dynamics, with preference probability $\sigma(\beta\Delta_\theta)$ and reference-adjusted margin Δ_θ. Together, these diagnostics show that TAB-PO concentrates optimization on schema-critical tokens while maintaining preferred-over-rejected separation.

TAB-PO is designed to correct mechanism-level failures that arise when preference optimization is applied to low-separation structured outputs: gradient dilution and preferred-token erosion. We therefore audit the training

dynamics directly, rather than relying only on downstream F1. The diagnostics separate completion tokens into three categories: *critical schema tokens*, which determine semantic labels, textual groundings, relation labels, relation arguments, sentence identifiers, or coreference links; *JSON scaffolding tokens*, which encode braces, brackets, commas, colons, and quotation marks; and *non-critical serialization tokens*, such as repeated field names or other formatting tokens that support parseability but are not themselves scored structured decisions.

Figure 3 reports three complementary diagnostics. Each diagnostic isolates a different part of the proposed mechanism: where the barrier activates, where gradient mass is allocated, and whether preference separation is preserved while vulnerable preferred tokens are restored.

1. **Barrier activity localizes likelihood restoration.** The barrier-activity panel measures the fraction of preferred tokens whose current policy probability falls below the confidence threshold τ and therefore activates the TAB-PO restoration term. TAB-PO activates most strongly on critical schema tokens, especially early in optimization, while remaining weak on non-critical serialization tokens. This shows that the confidence gate is not simply adding uniform SFT regularization; instead, it selectively restores the token positions where erosion would most affect structured correctness. JSON-scaffolding tokens receive weaker but nonzero activation, which is desirable when structural tokens must be protected to preserve machine-parseable outputs.

2. **Gradient mass shifts toward schema-critical decisions.** The gradient-mass panels examine where preference-optimization updates are allocated. Sequence-level and token-level DPO-family baselines allocate visible update mass to JSON scaffolding and other non-critical serialization tokens. In contrast, TAB-PO shifts a larger share of the learning signal toward critical schema tokens—the sparse positions that determine ontology labels, grounded spans, relation labels, and linking decisions. This directly supports the intended role of TAB-PO in reducing gradient dilution.

3. **Preference separation is preserved without token erosion.** The preference-dynamics panels track both the preference probability $\sigma(\beta\Delta_\theta)$ and the reference-adjusted margin Δ_θ. TAB-PO maintains strong preferred-over-rejected separation while the barrier remains active on under-confident

preferred tokens. Thus, the restoration term does not suppress preference learning; instead, it protects vulnerable preferred tokens while the model continues to increase the reference-adjusted preference margin.

Together, these diagnostics show that TAB-PO improves the allocation of optimization signal, preserves token-level stability, and maintains sequence-level preference separation.

8. Experimental Results

Scoring overview. Figure E1 of the Supplementary Material illustrates the two complementary structured prediction tasks evaluated in this paper, and Appendix E of the Supplementary Material provides the corresponding task-specific scoring definitions. PV-Miner evaluates hierarchical clinical communication mining through coarse Code F1 (parent-level semantic labelling), fine-grained Sub-code F1 (child-level semantic labelling), and grounded Span F1 (textual grounding). SciERC evaluates scientific information extraction through Entity F1 (entity semantic labelling), Relation F1 (typed relational linking), Span F1 (entity textual grounding), and Coreference F1 (cross-mention relational linking). Thus, the reported metrics separate semantic labelling, textual grounding, and relational linking quality.

We first verify that the modular prompt-engineered interface provides a stronger zero-shot initialization before supervised or preference-based fine-tuning. Appendix C of the Supplementary Material reports the full zero-shot comparison between baseline and prompt-engineered instructions in Table C1. Prompt engineering improves nearly all model–metric cells on PV-Miner and all cells on SciERC, supporting its role as an effective structured prediction interface before SFT and TAB-PO.

8.1. TAB-PO Consistently Improves over SFT

Table 1 reports the main post-SFT preference optimization results. On PV-Miner, TAB-PO improves over the corresponding SFT baseline by an average of +3.84 F1 across 18 model–metric cells, with the largest gains on the semantic hierarchy: +4.49 Code F1 and +6.25 Sub-code F1 on average.

On SciERC, the improvements are larger. TAB-PO improves over SFT by an average of +5.77 F1 across 24 model–metric cells, with gains concentrated on the most structurally demanding dimensions: +7.93 Relation F1 and +10.13 Coreference F1 on average. Entity F1 and Span F1 also improve by +1.22 and +3.80, respectively.

Table 1: F1 results of TAB-PO compared with recent sequence-level and token-level DPO-family methods on PV-Miner and SciERC. Parentheses report the change from the corresponding SFT baseline for the same model and task. Green cells indicate improvements and red cells indicate degradations; darker shading indicates a larger absolute change. All the results are the mean over 3 runs with different random seeds.

(a) PV-Miner

Model	Method	Code	Sub-code	Span
Llama-3.1 8B	Cal-DPO	77.43 (↓ 1.67)	74.51 (↑ 0.05)	86.83 (↓ 0.11)
	DPO	79.35 (↑ 0.25)	75.81 (↑ 1.35)	87.24 (↑ 0.30)
	DPOP	79.43 (↑ 0.33)	75.59 (↑ 1.13)	87.28 (↑ 0.34)
	IPO	12.75 (↓ 66.35)	6.25 (↓ 68.21)	6.00 (↓ 80.94)
	TDPO	77.89 (↓ 1.21)	73.77 (↓ 0.69)	85.38 (↓ 1.56)
	TI-DPO	79.80 (↑ 0.70)	75.75 (↑ 1.29)	87.18 (↑ 0.24)
	TAB-PO	82.99 (↑ 3.89)	80.62 (↑ 6.16)	87.67 (↑ 0.73)
Llama-3.2 3B	Cal-DPO	65.51 (↓ 9.61)	65.90 (↓ 3.74)	81.46 (↓ 2.29)
	DPO	74.80 (↓ 0.32)	70.68 (↑ 1.04)	84.48 (↑ 0.73)
	DPOP	74.89 (↓ 0.23)	71.08 (↑ 1.44)	84.25 (↑ 0.50)
	IPO	52.78 (↓ 22.34)	42.69 (↓ 26.95)	49.80 (↓ 33.95)
	TDPO	72.61 (↓ 2.51)	70.14 (↑ 0.50)	82.72 (↓ 1.03)
	TI-DPO	74.88 (↓ 0.24)	70.52 (↑ 0.88)	84.25 (↑ 0.50)
	TAB-PO	80.49 (↑ 5.37)	78.08 (↑ 8.44)	85.29 (↑ 1.54)
Llama-3.3 70B	Cal-DPO	79.77 (↓ 3.92)	75.49 (↓ 4.64)	86.21 (↓ 2.38)
	DPO	83.40 (↓ 0.29)	79.77 (↓ 0.36)	88.60 (↑ 0.01)
	DPOP	83.63 (↓ 0.06)	80.55 (↑ 0.42)	88.40 (↓ 0.19)
	IPO	8.52 (↓ 75.17)	3.21 (↓ 76.92)	11.02 (↓ 77.57)
	TDPO	83.33 (↓ 0.36)	79.70 (↓ 0.43)	88.44 (↓ 0.15)
	TI-DPO	83.67 (↓ 0.02)	80.38 (↑ 0.25)	88.69 (↑ 0.10)
	TAB-PO	84.97 (↑ 1.28)	82.71 (↑ 2.58)	88.03 (↓ 0.56)
Qwen2.5 1.5B	Cal-DPO	65.84 (↓ 7.56)	64.74 (↓ 2.17)	81.97 (↓ 2.32)
	DPO	68.30 (↓ 5.10)	65.36 (↓ 1.55)	84.61 (↑ 0.32)
	DPOP	69.78 (↓ 3.62)	66.66 (↓ 0.25)	84.17 (↓ 0.12)
	IPO	61.43 (↓ 11.97)	51.47 (↓ 15.44)	75.77 (↓ 8.52)
	TDPO	67.74 (↓ 5.66)	64.49 (↓ 2.42)	83.12 (↓ 1.17)
	TI-DPO	72.47 (↓ 0.93)	66.70 (↓ 0.21)	84.63 (↑ 0.34)
	TAB-PO	78.76 (↑ 5.36)	74.01 (↑ 7.10)	85.35 (↑ 1.06)
Qwen2.5 7B	Cal-DPO	74.67 (↓ 2.72)	71.82 (↓ 1.67)	85.64 (↓ 1.31)
	DPO	75.69 (↓ 1.70)	71.62 (↓ 1.87)	85.95 (↓ 1.00)
	DPOP	75.84 (↓ 1.55)	71.99 (↓ 1.50)	86.10 (↓ 0.85)
	IPO	43.44 (↓ 33.95)	32.21 (↓ 41.28)	30.92 (↓ 56.03)
	TDPO	76.01 (↓ 1.38)	71.76 (↓ 1.73)	86.10 (↓ 0.85)
	TI-DPO	76.84 (↓ 0.55)	71.91 (↓ 1.58)	86.18 (↓ 0.77)
	TAB-PO	81.86 (↑ 4.47)	78.31 (↑ 4.82)	87.32 (↑ 0.37)
Qwen2.5 14B	Cal-DPO	77.82 (↑ 2.13)	75.11 (↑ 4.00)	86.91 (↑ 0.95)
	DPO	79.06 (↑ 3.37)	74.88 (↑ 3.77)	87.07 (↑ 1.11)
	DPOP	79.11 (↑ 3.42)	74.91 (↑ 3.80)	87.05 (↑ 1.09)
	IPO	31.86 (↓ 43.83)	33.77 (↓ 37.34)	19.70 (↓ 66.26)
	TDPO	79.12 (↑ 3.43)	74.99 (↑ 3.88)	86.92 (↑ 0.96)
	TI-DPO	79.11 (↑ 3.42)	74.99 (↑ 3.88)	87.11 (↑ 1.15)
	TAB-PO	82.26 (↑ 6.57)	79.50 (↑ 8.39)	87.56 (↑ 1.60)

(b) SciERC

Model	Method	Entity	Relation	Span	Coreference
Llama-3.1 8B	Cal-DPO	90.73 (↑ 0.09)	50.70 (↑ 0.15)	85.38 (↓ 0.02)	41.23 (↑ 0.99)
	DPO	90.93 (↑ 0.29)	51.06 (↑ 0.51)	86.01 (↑ 0.61)	40.64 (↑ 0.40)
	DPOP	91.24 (↑ 0.60)	51.04 (↑ 0.49)	85.91 (↑ 0.51)	41.06 (↑ 0.82)
	IPO	90.50 (↓ 0.14)	50.50 (↓ 0.05)	85.89 (↑ 0.49)	40.91 (↑ 0.67)
	TDPO	90.83 (↑ 0.19)	51.12 (↑ 0.57)	85.94 (↑ 0.54)	39.26 (↓ 0.98)
	TI-DPO	90.32 (↓ 0.32)	49.54 (↓ 1.01)	85.85 (↑ 0.45)	41.56 (↑ 1.32)
	TAB-PO	91.50 (↑ 0.86)	56.63 (↑ 6.08)	86.31 (↑ 0.91)	46.24 (↑ 6.00)
Llama-3.2 3B	Cal-DPO	90.49 (↑ 0.65)	39.19 (↑ 0.54)	82.14 (↓ 1.11)	27.69 (↑ 0.55)
	DPO	90.12 (↑ 0.28)	39.56 (↑ 0.91)	83.36 (↑ 0.11)	30.98 (↑ 3.84)
	DPOP	90.36 (↑ 0.52)	39.81 (↑ 1.16)	83.42 (↑ 0.17)	28.45 (↑ 1.31)
	IPO	90.45 (↑ 0.61)	39.23 (↑ 0.58)	82.94 (↓ 0.31)	30.23 (↑ 3.09)
	TDPO	89.81 (↓ 0.03)	40.88 (↑ 2.23)	83.13 (↓ 0.12)	29.58 (↑ 2.44)
	TI-DPO	89.59 (↓ 0.25)	35.39 (↓ 3.26)	83.21 (↓ 0.04)	26.64 (↓ 0.50)
	TAB-PO	91.04 (↑ 1.20)	46.67 (↑ 8.02)	84.98 (↑ 1.73)	35.11 (↑ 7.97)
Llama-3.3 70B	Cal-DPO	90.83 (↓ 0.12)	54.55 (↑ 1.24)	86.24 (↓ 0.09)	49.72 (↑ 0.14)
	DPO	91.69 (↑ 0.74)	54.51 (↑ 1.20)	86.26 (↓ 0.07)	50.64 (↑ 1.06)
	DPOP	91.22 (↑ 0.27)	52.83 (↓ 0.48)	85.98 (↓ 0.35)	48.40 (↓ 1.18)
	IPO	90.65 (↓ 0.30)	54.34 (↑ 1.03)	86.07 (↓ 0.26)	50.24 (↑ 0.66)
	TDPO	90.99 (↑ 0.04)	53.74 (↑ 0.43)	87.02 (↑ 0.69)	46.72 (↓ 2.86)
	TI-DPO	89.00 (↓ 1.95)	53.82 (↑ 0.51)	86.80 (↑ 0.47)	45.13 (↓ 4.45)
	TAB-PO	92.21 (↑ 1.26)	58.68 (↑ 5.37)	88.11 (↑ 1.78)	60.80 (↑ 11.22)
Qwen2.5 1.5B	Cal-DPO	89.25 (↓ 0.02)	24.11 (↑ 1.60)	67.14 (↓ 0.33)	11.88 (↑ 2.15)
	DPO	89.32 (↑ 0.05)	25.16 (↑ 2.65)	68.85 (↑ 1.38)	12.05 (↑ 2.32)
	DPOP	88.41 (↓ 0.86)	26.95 (↑ 4.44)	70.83 (↑ 3.36)	13.12 (↑ 3.39)
	IPO	89.46 (↑ 0.19)	23.96 (↑ 1.45)	67.59 (↑ 0.12)	11.09 (↑ 1.36)
	TDPO	89.03 (↓ 0.24)	25.38 (↑ 2.87)	68.56 (↑ 1.09)	13.28 (↑ 3.55)
	TI-DPO	89.22 (↓ 0.05)	23.22 (↑ 0.71)	73.54 (↑ 6.07)	10.89 (↑ 1.16)
	TAB-PO	90.85 (↑ 1.58)	38.31 (↑ 15.80)	83.07 (↑ 15.60)	24.76 (↑ 15.03)
Qwen2.5 7B	Cal-DPO	90.21 (↑ 0.63)	41.23 (↑ 0.64)	82.62 (↑ 1.43)	29.99 (↓ 1.28)
	DPO	89.70 (↑ 0.12)	40.40 (↓ 0.19)	81.08 (↓ 0.11)	31.63 (↑ 0.36)
	DPOP	90.23 (↑ 0.65)	42.79 (↑ 2.20)	81.63 (↑ 0.44)	35.66 (↑ 4.39)
	IPO	89.99 (↑ 0.41)	41.21 (↑ 0.62)	80.97 (↓ 0.22)	32.25 (↑ 0.98)
	TDPO	89.60 (↑ 0.02)	40.63 (↑ 0.04)	80.76 (↓ 0.43)	33.68 (↑ 2.41)
	TI-DPO	89.50 (↓ 0.08)	42.97 (↑ 2.38)	83.42 (↑ 2.23)	28.47 (↓ 2.80)
	TAB-PO	91.50 (↑ 1.92)	48.88 (↑ 8.29)	83.38 (↑ 2.19)	41.19 (↑ 9.92)
Qwen2.5 14B	Cal-DPO	91.30 (↑ 0.17)	50.65 (↓ 0.45)	85.31 (↑ 0.70)	45.01 (↑ 2.67)
	DPO	91.07 (↓ 0.06)	49.77 (↓ 1.33)	85.43 (↑ 0.82)	43.41 (↑ 1.07)
	DPOP	90.67 (↓ 0.46)	51.76 (↑ 0.66)	85.48 (↑ 0.87)	46.49 (↑ 4.15)
	IPO	90.85 (↓ 0.28)	50.78 (↓ 0.32)	85.38 (↑ 0.77)	45.49 (↑ 3.15)
	TDPO	91.05 (↓ 0.08)	49.15 (↓ 1.95)	84.60 (↓ 0.01)	46.31 (↑ 3.97)
	TI-DPO	90.93 (↓ 0.20)	50.92 (↓ 0.18)	84.67 (↑ 0.06)	43.60 (↑ 1.26)
	TAB-PO	91.64 (↑ 0.51)	55.14 (↑ 4.04)	85.20 (↑ 0.59)	52.95 (↑ 10.61)

8.2. TAB-PO Outperforms Sequence-Level and Token-Level DPO Variants

TAB-PO also outperforms recent sequence-level and token-level DPO-family baselines. Against the strongest non-TAB-PO baseline for each model–metric cell, TAB-PO wins 39 of 42 comparisons overall: 17 of 18 on PV-Miner and 22 of 24 on SciERC. Notably, TAB-PO wins all semantic-labelling and relational-linking comparisons, covering PV-Miner Code/Sub-code and SciERC Entity/Relation/Coreference metrics.

The advantage is largest on token-critical structured decisions. On PV-

Miner, TAB-PO exceeds the best non-TAB-PO baseline by +4.09 Code F1 and +5.33 Sub-code F1 on average. On SciERC, it exceeds the best non-TAB-PO baseline by +6.01 Relation F1 and +7.07 Coreference F1 on average. These results support the central design of TAB-PO.

8.3. TAB-PO Surpasses Frontier Proprietary Models on SciERC

Because PV-Miner is privacy-constrained, frontier model comparisons are conducted only on SciERC. Figure 2 shows that TAB-PO 70B surpasses the strongest frontier proprietary model on every SciERC metric: +2.44 Entity F1, +8.37 Relation F1, +4.68 Span F1, and +16.33 Coreference F1.

The frontier comparison also shows that TAB-PO gains are not limited to the 70B model. Averaging Entity, Relation, Span, and Coreference F1, TAB-PO Qwen2.5-14B reaches 71.23 macro F1, exceeding every individual frontier baseline and the 67.00 composite formed from the best frontier score on each metric by 4.24 F1 points. TAB-PO Llama-3.2-3B reaches 64.45 macro F1, exceeding nine of the 11 frontier models. These results show gains for smaller open-weight models as well as the 70B model.

8.4. Ablation Study

We provide an empirical stress test of TAB-PO's proposed mechanism through targeted ablations. We separately evaluate its two central design choices: confusion-aware preference construction and confidence-gated likelihood restoration. We then analyze sensitivity to the number of rejected outputs per input, the type of preference construction, the type of confusion-aware perturbation, and key optimization hyperparameters, including the preference temperature, barrier confidence threshold, and barrier strength. Together, these ablations assess whether TAB-PO's gains arise from its intended mechanism rather than from incidental negative-sampling choices or hyperparameter effects.

Table 2 analyzes the components of TAB-PO. Increasing the number of rejected outputs per input example improves PV-Miner by +1.51 F1 with two negatives and +3.12 F1 with three negatives on average. On SciERC, the corresponding gains are smaller but consistently positive, with +0.62 F1 for two negatives and +0.55 F1 for three negatives. This suggests that additional hard negatives can improve coverage, but the central gains do not come merely from adding more preference pairs.

The confidence-gated barrier is essential. Lowering the threshold to $\tau = 0.25$ causes large average drops of -18.30 F1 on PV-Miner and -4.29 F1 on

Table 2: Diagnostic ablations for TAB-PO on PV-Miner and SciERC. For PV-Miner, the reference configuration uses one rejected output per input example, Code and Sub-code semantic-label confusions, confusion-aware hard-negative construction with expert-preferred chosen outputs, $\tau = 0.87$, $\lambda = 0.5$, and $\beta = 0.5$. For SciERC, the reference configuration uses one rejected output per input example, Entity and Relation semantic-label/relational-linking confusions, confusion-aware hard-negative construction with expert-preferred chosen outputs, $\tau = 0.87$, $\lambda = 0.5$, and $\beta = 0.5$. Parentheses report absolute F1 change from the corresponding task-specific reference configuration. Green cells indicate improvement and red cells indicate degradation; darker shading indicates larger relative change. All ablations use Llama-3.2-3B.

(a) PV-Miner

Diagnostic axis	Setting	Code	Sub-code	Span
Reference	One rejected output per input example; Code+Sub-code confusion; confusion-aware+ expert-preferred; $\tau = 0.87$, $\lambda = 0.5$, $\beta = 0.5$	80.49	78.08	85.29
A. Preference data preparation diagnostics				
Number of rejected outputs per input example	2	82.42 (↑ 1.93)	79.89 (↑ 1.81)	86.07 (↑ 0.78)
	3	**83.64** (↑ 3.15)	**82.60** (↑ 4.52)	**86.97** (↑ 1.68)
Preference construction	Expert-preferred rejected output	79.24 (↓ 1.25)	76.94 (↓ 1.14)	85.44 (↑ 0.15)
Confusion-aware preference construction	Extra record	80.01 (↓ 0.48)	77.88 (↓ 0.20)	84.59 (↓ 0.70)
	Missing record	80.82 (↑ 0.33)	77.77 (↓ 0.31)	86.50 (↑ 1.21)
B. TAB-PO objective diagnostics				
Barrier confidence threshold	$\tau = 0.25$	61.58 (↓ 18.91)	48.58 (↓ 29.50)	78.81 (↓ 6.48)
	$\tau = 0.50$	78.50 (↓ 1.99)	76.45 (↓ 1.63)	85.01 (↓ 0.28)
	$\tau = 0.95$	80.31 (↓ 0.18)	77.90 (↓ 0.18)	85.72 (↑ 0.43)
Barrier weight	$\lambda = 0.1$	79.13 (↓ 1.36)	76.99 (↓ 1.09)	84.93 (↓ 0.36)
	$\lambda = 2.0$	81.55 (↑ 1.06)	80.81 (↑ 2.73)	86.63 (↑ 1.34)
Preference temperature	$\beta = 0.25$	81.07 (↑ 0.58)	78.11 (↑ 0.03)	85.67 (↑ 0.38)
	$\beta = 0.75$	79.85 (↓ 0.64)	78.33 (↑ 0.25)	86.21 (↑ 0.92)

(b) SciERC

Diagnostic axis	Setting	Entity	Relation	Span	Coreference
Reference	One rejected output per input example; Entity+Relation confusion; confusion-aware+ expert-preferred; $\tau = 0.87$, $\lambda = 0.5$, $\beta = 0.5$	91.04	46.67	84.98	35.11
A. Preference data preparation diagnostics					
Number of rejected outputs per input example	2	**91.41** (↑ 0.37)	46.90 (↑ 0.23)	84.94 (↓ 0.04)	**37.04** (↑ 1.93)
	3	91.37 (↑ 0.33)	**47.24** (↑ 0.57)	85.33 (↑ 0.35)	36.06 (↑ 0.95)
Preference construction	Expert-preferred rejected output	90.79 (↓ 0.25)	44.29 (↓ 2.38)	84.14 (↓ 0.84)	33.54 (↓ 1.57)
Confusion-aware pair generation	Missing entity	90.81 (↓ 0.23)	46.80 (↑ 0.13)	**85.93** (↑ 0.95)	34.98 (↓ 0.13)
	Extra entity	90.72 (↓ 0.32)	44.89 (↓ 1.78)	84.45 (↓ 0.53)	34.14 (↓ 0.97)
	Extra relation	90.99 (↓ 0.05)	45.65 (↓ 1.02)	84.91 (↓ 0.07)	33.51 (↓ 1.60)
	Extra coreference	90.74 (↓ 0.30)	45.18 (↓ 1.49)	84.23 (↓ 0.75)	33.92 (↓ 1.19)
B. TAB-PO objective diagnostics					
Barrier confidence threshold	$\tau = 0.25$	90.05 (↓ 0.99)	43.13 (↓ 3.54)	83.76 (↓ 1.22)	23.71 (↓ 11.40)
	$\tau = 0.50$	90.31 (↓ 0.73)	43.98 (↓ 2.69)	83.88 (↓ 1.10)	26.87 (↓ 8.24)
	$\tau = 0.95$	90.79 (↓ 0.25)	46.33 (↓ 0.34)	84.62 (↓ 0.36)	34.70 (↓ 0.41)
Barrier weight	$\lambda = 0.1$	90.55 (↓ 0.49)	45.04 (↓ 1.63)	84.64 (↓ 0.34)	33.23 (↓ 1.88)
	$\lambda = 2.0$	91.21 (↑ 0.17)	46.80 (↑ 0.13)	84.50 (↓ 0.48)	35.41 (↑ 0.30)
Preference temperature	$\beta = 0.25$	91.03 (↓ 0.01)	46.31 (↓ 0.36)	85.10 (↑ 0.12)	35.28 (↑ 0.17)
	$\beta = 0.75$	90.34 (↓ 0.70)	45.80 (↓ 0.87)	84.20 (↓ 0.78)	34.73 (↓ 0.38)

SciERC, while $\tau = 0.50$ also degrades performance. A weak barrier ($\lambda = 0.1$) reduces performance on both tasks, with average drops of -0.94 F1 on PV-Miner and -1.09 F1 on SciERC. These results show that the barrier must activate on the appropriate under-confident preferred tokens with sufficient strength. Conversely, moderate changes to the preference temperature β produce smaller effects, indicating that TAB-PO is less sensitive to the margin temperature than to the confidence-gated anchoring mechanism.

The preference-construction diagnostics further rule out a simple data-only explanation. Replacing the default confusion-aware construction with expert-preferred negative construction reduces average performance by -0.75 F1 on PV-Miner and -1.26 F1 on SciERC. Extra-record and missing-record perturbations produce mixed effects, whereas the full confusion-aware

construction provides the strongest and most stable reference configuration. These ablations support the intended mechanism: TAB-PO works because it combines realistic ontology-level hard negatives with a token-level barrier that selectively protects under-confident preferred schema tokens.

Additional auxiliary loss components for structured prediction, including token weighting, length normalization, and class-balanced reweighting, are described in Appendix B of the Supplementary Material. These components address recurring challenges in structured prediction, including sparse task-critical tokens, variable-length evidence spans, and highly imbalanced ontology label distribution.

9. Discussion

TAB-PO is designed for settings in which preferred and rejected outputs are low-separation structured objects, and where correctness depends on a sparse set of schema-defining tokens. This makes the method especially suitable for ontology-driven structured prediction, a regime we evaluate through two complementary task settings: clinical communication mining in PV-Miner and scientific information extraction in SciERC. These tasks differ substantially in domain, ontology structure, and output dependencies, yet both expose the same underlying optimization challenge: small token-level changes can determine semantic labels, textual grounding, and relational links. The consistency of TAB-PO's gains across these settings suggests that its mechanism is not tied to a single ontology or dataset format, but to a broader class of token-critical structured prediction problems.

A natural direction for future work is to extend TAB-PO beyond structured information extraction to decision-oriented LLM systems in which a small number of tokens can alter downstream actions. Relevant settings include clinical safety triage, legal and regulatory compliance review, insurance claim assessment, quality auditing, decision-support workflows, and tool-calling systems. In such settings, the same low-separation preference geometry can arise: two outputs may be nearly identical in surface form but differ in a decisive label, action, rule interpretation, risk category, or tool argument. Extending TAB-PO to these domains would test whether confusion-aware preference construction and confidence-gated likelihood restoration can improve not only structured prediction accuracy, but also the reliability of LLM-assisted decision-making systems in high-stakes applications.

10. Conclusion

TAB-PO demonstrates that effective preference optimization for structured generation requires objectives and preference pairs that reflect and leverage the geometry of structured correctness. It addresses this setting through two coordinated components: confusion-aware preference construction, which creates schema-valid hard negatives from realistic SFT ontology-level error patterns together with expert-guided ambiguity analysis, and a confidence-gated token barrier, which protects under-confident preferred tokens from likelihood erosion while reducing gradient dilution across non-critical serialization tokens. These mechanisms translate into consistent gains on the dimensions most aligned with token-critical structured decisions. On SciERC, all six model configurations improve both Relation and Coreference F1 over their SFT baselines, with even the smallest gains across models still amounting to +4.04 and +6.00 points, respectively. The benefit holds across model scales from 1.5B to 70B parameters: Qwen2.5-1.5B improves SciERC macro F1 by +12.00 points after TAB-PO, while Llama-3.3-70B improves by +4.91 points. These results show that TAB-PO is a strong fit for ontology-driven structured prediction.

Statements and Declarations

Acknowledgements

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Ethics approval

This study used de-identified patient–provider secure-message data obtained from participating clinical sites, including Yale New Haven Health

System, Cleveland Clinic, and the Texas Association of Charitable Clinics, through approved institutional governance procedures. All procedures were approved by the Yale University IRB and conducted in accordance with relevant laws, institutional policies, and ethical guidelines for the analysis of de-identified clinical data. The project was reviewed under the applicable institutional governance process and determined not to constitute human-subjects research. The privacy rights of individuals were strictly observed, and no directly identifying patient information is reported in the manuscript.

Data and code availability

The SciERC dataset used in this study is publicly available from its original data source. The experiments on SciERC can be reproduced using the public SciERC data together with our publicly available code. The PV-Miner clinical dataset generated and analyzed during the current study is not publicly available because it is derived from privacy-protected patient–provider secure-message data obtained from Yale New Haven Health System, Cleveland Clinic, and the Texas Association of Charitable Clinics. These data are subject to strict institutional governance, IRB restrictions, data-use agreements, and privacy-protection obligations, and therefore cannot be shared publicly. The code developed for supervised fine-tuning, confusion-aware preference construction, TAB-PO optimization, inference, and metric evaluation on SciERC is publicly available at https://github.com/Data-Mining-Lab-Yale/TAB-PO.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Samah Fodeh: Methodology, Supervision, Project administration, Funding acquisition, Resources, Writing – review & editing. **Linhai Ma:** Conceptualization, Methodology, Software, Formal analysis, Visualization, Investigation, Writing – original draft. **Ganesh Puthiaraju:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Visualization, Writing – original draft. **Srivani Talakokkul:** Data curation, Writing – review & editing. **Afshan Khan:** Data curation, Visualization, Writing – review & editing. **Sreeraj Ramachandran:** Software, Validation, Writing – review & editing. **Elyas Irankhah:** Software, Validation, Writing – review &

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